

CareerMatch AI – AI/ML Interview Preparation Guide

Generalized Educational Knowledge – Hackathon Prototype

Introduction

This guide gives CareerMatch AI structured, retrievable content for generating personalized interview preparation across 30 core AI/ML topics, plus a large bank of practice questions. This is generalized educational content for a hackathon prototype.

Topic-by-Topic Interview Preparation

1. Python Interview Topics

What the interviewer wants to know: Whether the candidate can write clean, correct, working code, not just recite syntax.

Beginner explanation: Understands core Python: data types, loops, functions, comprehensions, and common built-in structures.

Important concepts:

- Lists, tuples, dictionaries, and sets and when to use each
- List/dict comprehensions
- Functions, *args/**kwargs
- Basic time complexity of common operations

Common interview questions:

- 1 What is the difference between a list and a tuple?
- 2 How does a Python dictionary achieve fast lookups?

Scenario-based question: Given a list of numbers, write a function to return only the duplicates.

Project-related question: Describe a Python script you wrote and why you chose the data structures you used.

Common mistake: Memorizing syntax without understanding when to apply which structure.

Preparation tip: Practice writing small functions from scratch instead of only reading solutions.

2. SQL Interview Topics

What the interviewer wants to know: Whether the candidate can retrieve and shape real data correctly under mild ambiguity.

Beginner explanation: Comfortable with SELECT, filtering, grouping, and combining multiple tables.

Important concepts:

- SELECT, WHERE, GROUP BY, HAVING
- JOIN types (inner, left, right)

- Aggregate functions
- Basic subqueries

Common interview questions:

- 1 What is the difference between INNER JOIN and LEFT JOIN?
- 2 How would you find duplicate rows in a table?

Scenario-based question: You have a sales table and a products table; find the top-selling product per month.

Project-related question: Walk through a query you wrote to answer a real business question.

Common mistake: Confusing WHERE and HAVING, or forgetting that HAVING filters after aggregation.

Preparation tip: Practice on real, messy sample databases rather than only clean textbook tables.

3. Data Structures Basics

What the interviewer wants to know: Whether the candidate understands how data is organized and the trade-offs between structures.

Beginner explanation: Understands arrays, linked lists, stacks, queues, and basic hash maps.

Important concepts:

- Array vs linked list trade-offs
- Stack (LIFO) and queue (FIFO) use cases
- Hash maps for fast lookup
- Basic trees and graphs

Common interview questions:

- 1 When would you use a stack instead of a queue?
- 2 How does a hash map achieve near constant-time lookup?

Scenario-based question: Design a data structure to track the most recent 10 user actions efficiently.

Project-related question: Describe a data structure choice you made in a project and why.

Common mistake: Choosing a data structure without considering the actual access pattern needed.

Preparation tip: Practice implementing structures from scratch, not just using built-in ones.

4. Statistics

What the interviewer wants to know: Whether the candidate can reason correctly about data rather than just compute numbers.

Beginner explanation: Understands mean, median, standard deviation, and basic distributions.

Important concepts:

- Mean vs median vs mode
- Standard deviation and variance
- Normal distribution
- Correlation vs causation

Common interview questions:

- 1 Why might median be preferred over mean for skewed data?
- 2 What does a high standard deviation tell you about a dataset?

Scenario-based question: A dataset has a few extreme outliers; how would that affect your summary statistics?

Project-related question: Explain a time statistics helped you avoid a wrong conclusion from data.

Common mistake: Assuming correlation implies causation without further evidence.

Preparation tip: Practice interpreting real datasets, not just calculating formulas.

5. Machine Learning

What the interviewer wants to know: Whether the candidate understands ML at a conceptual level, not just how to call a library function.

Beginner explanation: Understands what it means for a model to 'learn' from data and generalize to new data.

Important concepts:

- Supervised vs unsupervised learning
- Training vs test data
- Generalization vs memorization
- Bias-variance trade-off

Common interview questions:

- 1 What does it mean for a model to generalize well?
- 2 Why do we split data into training and test sets?

Scenario-based question: Your model performs perfectly on training data but poorly on new data. What's happening?

Project-related question: Walk through the full pipeline of a machine learning project you built.

Common mistake: Evaluating a model only on training data.

Preparation tip: Build at least one complete project end-to-end rather than only following tutorials.

6. Supervised Learning

What the interviewer wants to know: Whether the candidate understands learning from labeled data and can pick suitable methods.

Beginner explanation: Understands that supervised learning uses input-output pairs to learn a mapping.

Important concepts:

- Labeled data requirement
- Regression vs classification as supervised sub-types
- Loss functions

Common interview questions:

- 1 What makes a learning problem 'supervised'?
- 2 Give an example of a real-world supervised learning problem.

Scenario-based question: You only have partially labeled data. How would you approach the problem?

Project-related question: Describe a supervised learning project and what the labels represented.

Common mistake: Confusing supervised learning with any task that simply uses data.

Preparation tip: Be ready to name the label/target variable in any project you discuss.

7. Unsupervised Learning

What the interviewer wants to know: Whether the candidate understands learning without labels and its practical uses.

Beginner explanation: Understands that unsupervised learning finds structure in data without predefined labels.

Important concepts:

- Clustering as a common unsupervised task
- Dimensionality reduction concept
- When labels are unavailable or expensive

Common interview questions:

- 1 When would you use unsupervised learning instead of supervised learning?
- 2 What is clustering used for?

Scenario-based question: You want to segment customers but have no predefined categories. How would you proceed?

Project-related question: Describe a project where you grouped or explored data without labels.

Common mistake: Assuming unsupervised results are automatically meaningful without validation.

Preparation tip: Practice interpreting and sanity-checking cluster results, not just producing them.

8. Regression

What the interviewer wants to know: Whether the candidate can predict continuous values and interpret model output correctly.

Beginner explanation: Understands regression predicts a numeric value based on input features.

Important concepts:

- Linear regression fundamentals
- Interpreting coefficients
- Evaluating fit (R-squared, RMSE)

Common interview questions:

- 1 What does the coefficient of a linear regression model represent?
- 2 How would you evaluate whether a regression model fits well?

Scenario-based question: Your regression model has a high R-squared on training data but performs poorly live. Why might that be?

Project-related question: Walk through a regression project and how you validated the predictions.

Common mistake: Reporting R-squared alone without checking residuals or overfitting.

Preparation tip: Practice explaining regression coefficients in plain, non-technical language.

9. Classification

What the interviewer wants to know: Whether the candidate can distinguish categories correctly and choose fair evaluation metrics.

Beginner explanation: Understands classification assigns inputs to discrete categories.

Important concepts:

- Binary vs multi-class classification
- Decision boundaries
- Handling class imbalance

Common interview questions:

- 1 How would you handle a classification problem with 95% of data in one class?
- 2 What's the difference between binary and multi-class classification?

Scenario-based question: A fraud detection model has 99% accuracy but misses most fraud cases. What's wrong?

Project-related question: Describe a classification project and which metric mattered most and why.

Common mistake: Relying on accuracy alone for an imbalanced classification problem.

Preparation tip: Always ask what the cost of a false positive vs false negative is for the problem.

10. Overfitting

What the interviewer wants to know: Whether the candidate recognizes when a model has memorized rather than learned.

Beginner explanation: Understands overfitting happens when a model performs well on training data but poorly on new data.

Important concepts:

- Signs of overfitting
- Regularization as a fix
- Cross-validation to detect it early

Common interview questions:

- 1 What are signs that a model is overfitting?
- 2 Name two ways to reduce overfitting.

Scenario-based question: Your model's training accuracy keeps improving but validation accuracy is dropping. What do you do?

Project-related question: Describe a time you diagnosed and fixed overfitting in a project.

Common mistake: Adding more model complexity to fix poor performance without checking for overfitting first.

Preparation tip: Always compare training and validation performance together, not in isolation.

11. Underfitting

What the interviewer wants to know: Whether the candidate understands the opposite failure mode: a model too simple to learn patterns.

Beginner explanation: Understands underfitting happens when a model is too simple to capture the underlying pattern.

Important concepts:

- Signs of underfitting
- Increasing model complexity or features as a fix

Common interview questions:

- 1 How would you know if your model is underfitting?
- 2 What steps would you take to fix an underfitting model?

Scenario-based question: Both your training and test accuracy are low. What could be the cause?

Project-related question: Describe a case where a simple model was insufficient for the problem.

Common mistake: Assuming poor performance always means overfitting, without checking training performance too.

Preparation tip: Check both training and validation error before deciding what's wrong.

12. Feature Engineering

What the interviewer wants to know: Whether the candidate can improve model performance through thoughtful data preparation.

Beginner explanation: Understands that transforming raw data into better features often improves results more than model choice.

Important concepts:

- Encoding categorical variables
- Scaling numerical features
- Creating domain-specific derived features

Common interview questions:

- 1 Why might feature engineering matter more than model choice in some cases?
- 2 How would you encode a categorical variable with many categories?

Scenario-based question: You have a 'date of purchase' column; what features could you derive from it?

Project-related question: Describe a feature you engineered that meaningfully improved a model.

Common mistake: Blindly encoding all categorical variables the same way regardless of cardinality.

Preparation tip: Always tie feature ideas back to domain knowledge of the problem.

13. Model Evaluation

What the interviewer wants to know: Whether the candidate can judge model quality correctly and choose the right metric for the problem.

Beginner explanation: Understands that different problems require different evaluation metrics.

Important concepts:

- Train/test/validation splits
- Cross-validation
- Choosing metrics appropriate to the problem

Common interview questions:

- 1 Why is a train/test split necessary?
- 2 When would you use cross-validation instead of a single split?

Scenario-based question: You're evaluating a medical screening model; which metric would you prioritize and why?

Project-related question: Describe how you evaluated and selected the final model in a project.

Common mistake: Using the same data for both training and evaluation.

Preparation tip: Be ready to justify your chosen metric based on the real-world cost of errors.

14. Precision

What the interviewer wants to know: Whether the candidate understands what precision measures and when it matters most.

Beginner explanation: Precision measures, of everything predicted positive, how much was actually positive.

Important concepts:

- Precision formula: $TP / (TP + FP)$
- High-cost false positive scenarios

Common interview questions:

- 1 When does precision matter more than recall?
- 2 Give an example where a low-precision model would be harmful.

Scenario-based question: In a spam filter, why might high precision be prioritized over high recall?

Project-related question: Describe a project where precision was your key evaluation metric.

Common mistake: Confusing precision with overall accuracy.

Preparation tip: Memorize the formula alongside a concrete real-world example.

15. Recall

What the interviewer wants to know: Whether the candidate understands what recall measures and when missing positives is costly.

Beginner explanation: Recall measures, of all actual positives, how many the model correctly identified.

Important concepts:

- Recall formula: $TP / (TP + FN)$
- High-cost false negative scenarios

Common interview questions:

- 1 When does recall matter more than precision?
- 2 Give an example where a low-recall model would be dangerous.

Scenario-based question: In disease screening, why is recall often prioritized over precision?

Project-related question: Describe a project where you optimized specifically for recall.

Common mistake: Not connecting the metric choice back to real-world consequences.

Preparation tip: Practice explaining precision and recall together using one shared example.

16. F1 Score

What the interviewer wants to know: Whether the candidate can balance precision and recall using a single combined metric.

Beginner explanation: F1 score is the harmonic mean of precision and recall, useful when both matter.

Important concepts:

- F1 formula and why harmonic mean is used
- When F1 is preferred over accuracy

Common interview questions:

- 1 Why use the harmonic mean instead of a simple average for F1?
- 2 When would F1 score be more useful than accuracy?

Scenario-based question: Your classes are imbalanced and both false positives and false negatives are costly. What metric fits best?

Project-related question: Describe a case where F1 score changed your model selection decision.

Common mistake: Using F1 score without understanding what precision/recall trade-off it hides.

Preparation tip: Always report precision and recall alongside F1 for full context.

17. Confusion Matrix

What the interviewer wants to know: Whether the candidate can read and interpret a full breakdown of classification results.

Beginner explanation: A confusion matrix shows true positives, false positives, true negatives, and false negatives.

Important concepts:

- Structure of a confusion matrix
- Deriving precision, recall, and accuracy from it

Common interview questions:

- 1 How would you derive precision and recall from a confusion matrix?
- 2 What does a high false negative count indicate?

Scenario-based question: Given a confusion matrix, identify whether the model struggles more with false positives or false negatives.

Project-related question: Describe how a confusion matrix changed how you understood your model's errors.

Common mistake: Only looking at accuracy and skipping the confusion matrix entirely.

Preparation tip: Practice building and reading confusion matrices by hand on small examples.

18. Decision Trees

What the interviewer wants to know: Whether the candidate understands a simple, interpretable model and its weaknesses.

Beginner explanation: A decision tree splits data based on feature values to make predictions, forming a tree-like structure.

Important concepts:

- How splits are chosen
- Overfitting risk with deep trees
- Interpretability advantage

Common interview questions:

- 1 Why are decision trees prone to overfitting?
- 2 What is one advantage of decision trees over more complex models?

Scenario-based question: Your decision tree performs very well on training data but poorly on new data. What would you try?

Project-related question: Describe a project where you used or compared decision trees.

Common mistake: Letting a tree grow without limiting depth or pruning.

Preparation tip: Understand pruning and max-depth as basic ways to control complexity.

19. Random Forest

What the interviewer wants to know: Whether the candidate understands ensembling and why it improves on single trees.

Beginner explanation: A random forest combines many decision trees trained on random subsets of data to improve accuracy and reduce overfitting.

Important concepts:

- Ensemble learning concept
- Bagging and randomness in feature/data selection

Common interview questions:

- 1 How does a random forest reduce overfitting compared to a single decision tree?
- 2 What does 'bagging' mean in this context?

Scenario-based question: You need a robust baseline model quickly for a new dataset; why might random forest be a good choice?

Project-related question: Describe a project where random forest outperformed a simpler model.

Common mistake: Treating random forest as a black box without understanding why ensembling helps.

Preparation tip: Be ready to explain bagging in your own simple words.

20. Gradient Boosting

What the interviewer wants to know: Whether the candidate understands sequential ensembling and when it's worth the complexity.

Beginner explanation: Gradient boosting builds models sequentially, each correcting the errors of the previous one.

Important concepts:

- Difference from bagging (random forest)
- Risk of overfitting with too many boosting rounds

Common interview questions:

- 1 How does gradient boosting differ from random forest?
- 2 Why can gradient boosting overfit if not tuned carefully?

Scenario-based question: Your boosted model outperforms random forest on training data but not on validation data. What would you check?

Project-related question: Describe a scenario where you chose gradient boosting over a simpler model.

Common mistake: Not tuning learning rate or number of estimators, leading to overfitting.

Preparation tip: Understand the trade-off between number of boosting rounds and overfitting.

21. Deep Learning

What the interviewer wants to know: Whether the candidate understands when deep learning is appropriate versus classical ML.

Beginner explanation: Deep learning uses multi-layered neural networks to learn complex patterns, especially from unstructured data.

Important concepts:

- When deep learning outperforms classical ML
- Data and compute requirements
- Basic neural network structure

Common interview questions:

- 1 When would you choose deep learning over a classical ML model?
- 2 What is a common trade-off of using deep learning?

Scenario-based question: You have a small tabular dataset; would you recommend deep learning? Why or why not?

Project-related question: Describe a deep learning project and why that approach was necessary.

Common mistake: Defaulting to deep learning for every problem regardless of data size or type.

Preparation tip: Learn to justify model choice based on data type and size, not popularity.

22. CNN

What the interviewer wants to know: Whether the candidate understands how vision models process spatial data.

Beginner explanation: A Convolutional Neural Network uses filters to detect spatial patterns like edges and shapes in images.

Important concepts:

- Convolution and pooling layers
- Why CNNs suit image data better than plain neural networks

Common interview questions:

- 1 Why are CNNs preferred over fully connected networks for images?
- 2 What does a pooling layer do?

Scenario-based question: Your CNN performs well on training images but poorly on slightly rotated test images. What could help?

Project-related question: Describe an image-based project where you used a CNN.

Common mistake: Not using data augmentation when the training set is small.

Preparation tip: Learn the roles of convolution, pooling, and fully connected layers separately.

23. NLP

What the interviewer wants to know: Whether the candidate understands the basics of processing human language computationally.

Beginner explanation: NLP is the field of AI focused on understanding and generating human language.

Important concepts:

- Text preprocessing steps
- From classical NLP to modern transformer-based methods

Common interview questions:

- 1 What preprocessing steps are common in an NLP pipeline?
- 2 How has NLP changed with the introduction of transformers?

Scenario-based question: You need to classify customer complaints by topic. How would you approach it?

Project-related question: Describe an NLP project and the preprocessing steps you used.

Common mistake: Skipping text cleaning and expecting a model to handle raw noisy text well.

Preparation tip: Practice a full pipeline: clean text, represent it numerically, then model it.

24. Transformers

What the interviewer wants to know: Whether the candidate understands the architecture behind modern language models.

Beginner explanation: Transformers use an attention mechanism to weigh the importance of different words in a sequence.

Important concepts:

- Attention mechanism at a high level
- Why transformers replaced older sequence models like RNNs

Common interview questions:

- 1 What problem does the attention mechanism solve?
- 2 Why are transformers well suited to parallel training?

Scenario-based question: Why might a transformer-based model understand long sentences better than older models?

Project-related question: Describe a project where you used a pretrained transformer model.

Common mistake: Trying to explain transformers only in terms of RNNs instead of their own mechanism.

Preparation tip: Focus on understanding attention conceptually before diving into implementation details.

25. Generative AI

What the interviewer wants to know: Whether the candidate understands how generative systems differ from predictive ones.

Beginner explanation: Generative AI creates new content, such as text or images, rather than only predicting a label.

Important concepts:

- Generative vs discriminative models
- Common use cases and limitations

Common interview questions:

- 1 How does a generative model differ from a discriminative model?
- 2 What is one limitation of current generative AI systems?

Scenario-based question: You want to summarize customer feedback automatically; how would generative AI help?

Project-related question: Describe a generative AI application you have built or used meaningfully.

Common mistake: Assuming generative AI outputs are always factually correct.

Preparation tip: Learn to evaluate generative outputs critically, not just accept them at face value.

26. LLMs

What the interviewer wants to know: Whether the candidate understands what large language models are and their practical limitations.

Beginner explanation: LLMs are large transformer-based models trained on massive text data to understand and generate language.

Important concepts:

- Strengths of LLMs (fluency, general knowledge)
- Limitations (hallucination, outdated knowledge)

Common interview questions:

- 1 What is 'hallucination' in the context of LLMs?
- 2 Why might an LLM give an outdated or incorrect factual answer?

Scenario-based question: A user asks your LLM-based app a question about very recent events. What would you do?

Project-related question: Describe an application you built that integrates an LLM.

Common mistake: Trusting LLM outputs without any grounding or verification step.

Preparation tip: Learn why grounding techniques like RAG exist to address LLM limitations.

27. RAG

What the interviewer wants to know: Whether the candidate understands how to ground AI answers in real, retrievable data.

Beginner explanation: Retrieval-Augmented Generation retrieves relevant information before generating a response, improving factual accuracy.

Important concepts:

- Retrieve-then-generate pipeline
- Role of embeddings and a vector database in retrieval

Common interview questions:

- 1 Why does RAG reduce hallucination compared to a plain LLM?
- 2 What are the main components of a RAG pipeline?

Scenario-based question: Your RAG system retrieves irrelevant documents for some queries. What would you check first?

Project-related question: Describe the RAG pipeline you built, step by step.

Common mistake: Feeding retrieved documents to the model without checking their relevance first.

Preparation tip: Be ready to explain chunking, embedding, retrieval, and generation as separate steps.

28. Embeddings

What the interviewer wants to know: Whether the candidate understands how meaning is represented numerically for search and comparison.

Beginner explanation: Embeddings are vector representations of data that capture semantic meaning, allowing similarity comparison.

Important concepts:

- Why similar meanings produce similar vectors
- Common similarity measures such as cosine similarity

Common interview questions:

- 1 What does it mean for two embeddings to be 'close' to each other?
- 2 Why are embeddings useful for search?

Scenario-based question: Two differently worded resumes describe the same skill; how would embeddings help match them?

Project-related question: Describe a project where you used embeddings for similarity or search.

Common mistake: Assuming embeddings capture meaning perfectly regardless of domain.

Preparation tip: Understand at least one similarity metric, such as cosine similarity, in detail.

29. Vector Databases

What the interviewer wants to know: Whether the candidate understands the infrastructure that makes large-scale semantic search possible.

Beginner explanation: Vector databases store and search embeddings efficiently, even across millions of items.

Important concepts:

- Why traditional databases struggle with similarity search
- Approximate nearest neighbor search concept

Common interview questions:

- 1 Why can't a traditional relational database efficiently perform similarity search?
- 2 What is approximate nearest neighbor search?

Scenario-based question: Your knowledge base grows to millions of documents; how does this affect your retrieval design?

Project-related question: Describe how you structured and queried a vector database in a project.

Common mistake: Not considering index type trade-offs between speed and accuracy.

Preparation tip: Learn the basic difference between exact and approximate similarity search.

30. Model Deployment

What the interviewer wants to know: Whether the candidate understands what happens after a model is trained, not just how to train one.

Beginner explanation: Model deployment makes a trained model accessible for real use, typically through an API or application.

Important concepts:

- Wrapping a model in an API
- Basic monitoring after deployment
- Versioning models over time

Common interview questions:

- 1 What steps are involved in taking a model from a notebook to a usable application?
- 2 Why is monitoring important after deployment?

Scenario-based question: Your deployed model's accuracy silently degrades over a few months. What might be happening, and what would you do?

Project-related question: Describe how you deployed a model in a personal or academic project.

Common mistake: Treating training as the final step and never testing the deployed version end-to-end.

Preparation tip: Practice deploying at least one small model with FastAPI or Streamlit.

30 Beginner Questions

- 1 What is the difference between a list and a tuple in Python?
- 2 What is a dictionary in Python and when would you use one?
- 3 Write a SQL query to select all rows where a column value is greater than 100.
- 4 What is the difference between WHERE and HAVING in SQL?
- 5 What is the difference between supervised and unsupervised learning?
- 6 What is a training set and a test set?
- 7 What does 'overfitting' mean in simple terms?
- 8 What is the difference between mean and median?
- 9 What is a variable, in the statistical sense?
- 10 What is a confusion matrix?
- 11 What does 'accuracy' measure in a classification model?
- 12 What is the difference between classification and regression?
- 13 What is a neural network, in simple terms?
- 14 What does 'feature' mean in a machine learning dataset?
- 15 What is the purpose of data cleaning?
- 16 What is a NumPy array, and how is it different from a Python list?
- 17 What is Pandas used for?
- 18 What does it mean to 'normalize' data?
- 19 What is a for loop used for in Python?
- 20 What is the difference between a function and a method?
- 21 What is Git used for?
- 22 What is an API, in simple terms?
- 23 What is the difference between a relational and non-relational database?
- 24 What is exploratory data analysis (EDA)?
- 25 What does 'label' mean in a supervised learning dataset?
- 26 What is a JOIN in SQL used for?
- 27 What is the purpose of a validation set?
- 28 What is tokenization in NLP?
- 29 What is an embedding, in simple terms?
- 30 What is the difference between a chatbot and an LLM-based assistant?

20 Intermediate Questions

- 1 Explain the bias-variance trade-off.
- 2 How would you handle missing data in a dataset with multiple types of columns?
- 3 What is regularization, and why is it used?
- 4 Compare precision, recall, and F1 score, and explain when each matters most.
- 5 How does a random forest reduce the overfitting risk seen in single decision trees?
- 6 What is cross-validation, and why is it more reliable than a single train/test split?
- 7 Explain how gradient descent works at a conceptual level.
- 8 What is the curse of dimensionality?
- 9 How would you detect and handle an imbalanced classification dataset?
- 10 What is the difference between bagging and boosting?
- 11 Explain how a convolutional layer extracts features from an image.
- 12 What is transfer learning, and why is it useful with limited data?
- 13 How does attention in a transformer differ from processing in an RNN?
- 14 What is the difference between fine-tuning and prompt engineering for an LLM?
- 15 How would you evaluate the quality of a RAG system's answers?
- 16 What is the difference between exact and approximate nearest neighbor search?
- 17 How would you design a schema for a system with millions of daily transactions?
- 18 What is the difference between horizontal and vertical scaling?
- 19 How would you monitor a deployed machine learning model over time?
- 20 What is data leakage, and how would you prevent it?

10 Project Discussion Questions

- 1 Walk me through your most challenging project from start to finish.
- 2 What was the hardest bug or issue you faced in this project, and how did you resolve it?
- 3 Why did you choose this particular model or approach for the project?
- 4 What would you do differently if you rebuilt this project today?
- 5 How did you validate that your project's results were actually correct?
- 6 What real-world assumptions does your project rely on, and are they realistic?
- 7 How did you decide which features or data to include in your project?
- 8 What was the impact or intended use case of your project?
- 9 How did you handle a case where your project's data was incomplete or messy?
- 10 If you had more time and resources, how would you extend this project?

10 Scenario-Based Questions

- 1 Your model performs well in testing but poorly once deployed. How would you investigate why?
- 2 A stakeholder asks you to improve model accuracy at any cost. How would you respond?

- 3 You discover your training data contains a labeling error affecting 10% of records. What do you do?
- 4 Your RAG system keeps citing irrelevant sources for certain queries. How would you debug this?
- 5 You need to choose between a highly accurate but uninterpretable model and a simpler, explainable one for a healthcare use case. How do you decide?
- 6 Your data pipeline fails silently overnight and stale data is used for a day. How would you prevent this in the future?
- 7 A classification model shows high overall accuracy but consistently fails for one specific subgroup. How would you address this?
- 8 You have very limited labeled data for an important classification task. What approaches would you consider?
- 9 Your team disagrees on which evaluation metric to prioritize for a new model. How would you help resolve this?
- 10 You're asked to add a generative AI feature to an existing product under a tight deadline. How would you scope the work responsibly?

10 Python/SQL Coding Questions

- 1 Write a Python function to check if a given string is a palindrome.
- 2 Write a Python function to find the second-largest number in a list without using sort().
- 3 Write a SQL query to find the second-highest salary in an employee table.
- 4 Write a Python function that returns the frequency count of each word in a sentence.
- 5 Write a SQL query to find customers who have placed more than 3 orders.
- 6 Write a Python function to remove duplicates from a list while preserving order.
- 7 Write a SQL query using GROUP BY and HAVING to find departments with more than 5 employees.
- 8 Write a Python function to merge two sorted lists into one sorted list.
- 9 Write a SQL query to join three tables to produce a combined report.
- 10 Write a Python function that flattens a nested list of arbitrary depth.

Beginner-Friendly Interview Preparation Checklist

- Can explain at least two complete projects clearly, including the 'why' behind key decisions
- Comfortable with Python and SQL fundamentals without needing notes
- Can explain precision, recall, F1, and confusion matrix without hesitation
- Understands overfitting/underfitting and can describe how to detect and fix each
- Can explain at least one deep learning and one generative AI concept in plain language
- Has practiced answering at least 5 scenario-based questions out loud
- Has a short, clear answer ready for 'walk me through a project of yours'
- Knows the trade-offs of the tools and models used in their own projects
- Can write simple Python and SQL code without relying on an IDE's autocomplete
- Has questions prepared to ask the interviewer

RAG Retrieval Keywords

AI ML interview questions, machine learning interview prep, Python interview questions, SQL interview questions, data science interview, deep learning interview, generative AI interview questions, RAG interview questions, scenario based interview questions, coding interview questions

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